NTNU – Trondheim
Norwegian University of
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TET4185: Power Markets, Resources and Environment
Pricing

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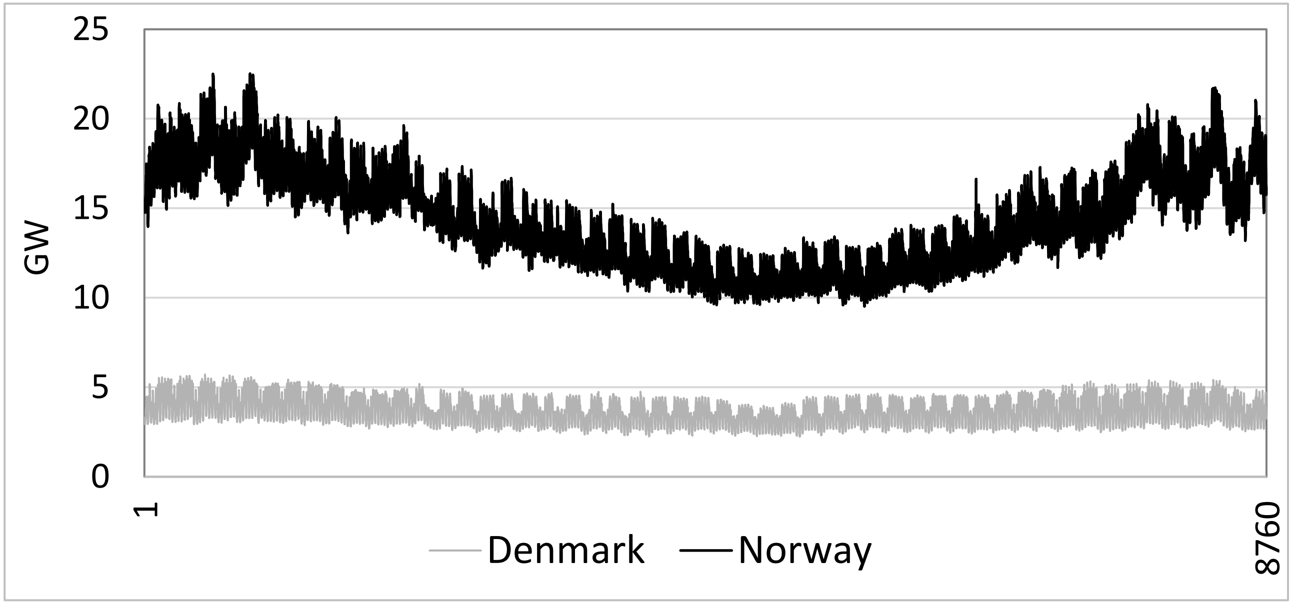
Overview

- Tariffs
- Restructuring and Unbundling
- Contracts
- Grid Tariffs
- Norwegian Grid Tariffs
- Flexible Demands

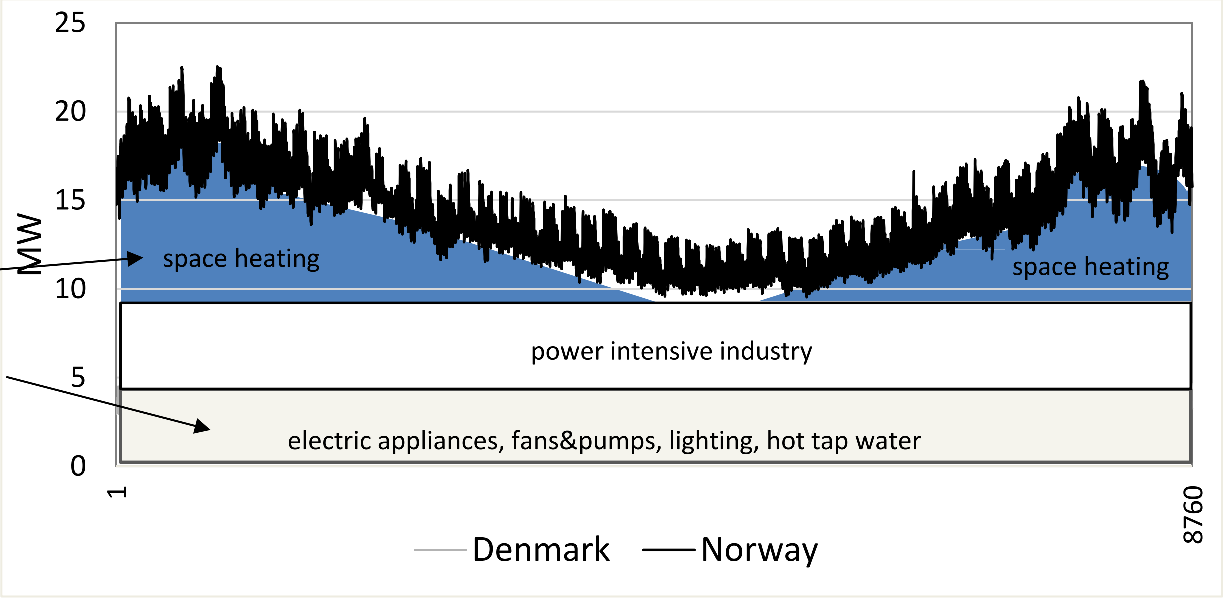
Electric load

- Norway 5.5 mill.
- Denmark 6 mill.



Composition of the electric load

- Industry
- Buildings
  - Heat load
  - Electric specific load
- Transport
  - EVs



Tariffs

- **Definition of Tariff:** In a traditionally structured power sector, all costs (e.g., operational, capital, maintenance) are bundled together into a single charge called a tariff. This approach simplifies billing for consumers but may not fully reflect individual cost components.

- **Major considerations:**
  - **Revenue adequacy:** Ensures that the tariff structure generates enough revenue to cover the costs of the power system, including generation, transmission, distribution, and maintenance.
  - **Economic efficiency:** Tariffs should incentivize optimal use of electricity, such as reducing consumption during peak hours or encouraging energy-saving behaviors.
  - **Distributional effects:** The impact of tariffs on different consumer groups (e.g., households, industries) should be fair and equitable. This includes addressing affordability for vulnerable consumers
  - **Simple and stable:** Tariffs need to be easy to understand and predictable over time, helping consumers plan their electricity use and budgets effectively.

Restructuring and Unbundling

- Unbundling: splitting up of the power sector in
  - Generation
  - Transmission and Distribution
  - Retail

- Unbundling leads to splitting up of the tariff into:
  - **Contract (power price):** Energy cost paid to generators or suppliers
  - **Grid tariff:** Covers costs for transmission and distribution

Contracts and Tariffs

- In a well-functioning market, power prices are determined by <u>market</u>
  - This pricing model allows prices to reflect scarcity, generation costs, and consumer preferences.

- The <u>grid tariff</u> has the same requirements as before:
  - Revenue adequacy
  - Economic efficiency
  - Distributional effects
  - Simple and stable
  - Acceptability

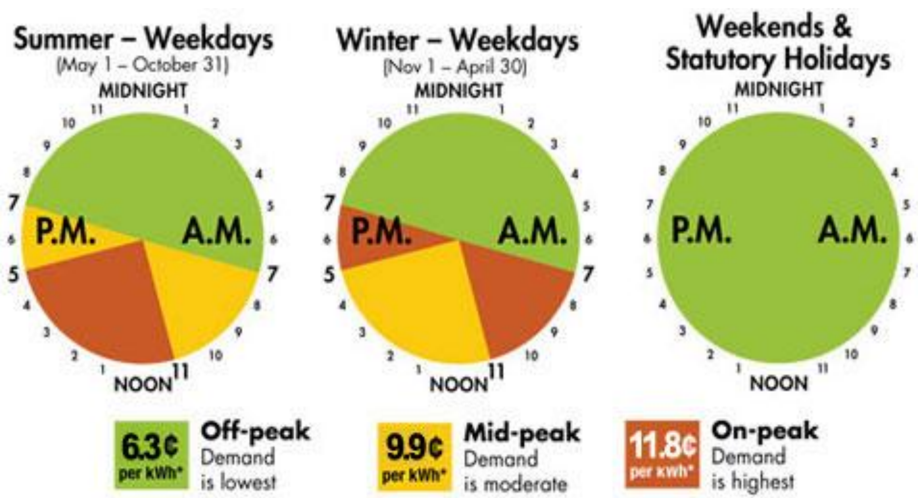
CONTRACTS

Contract types

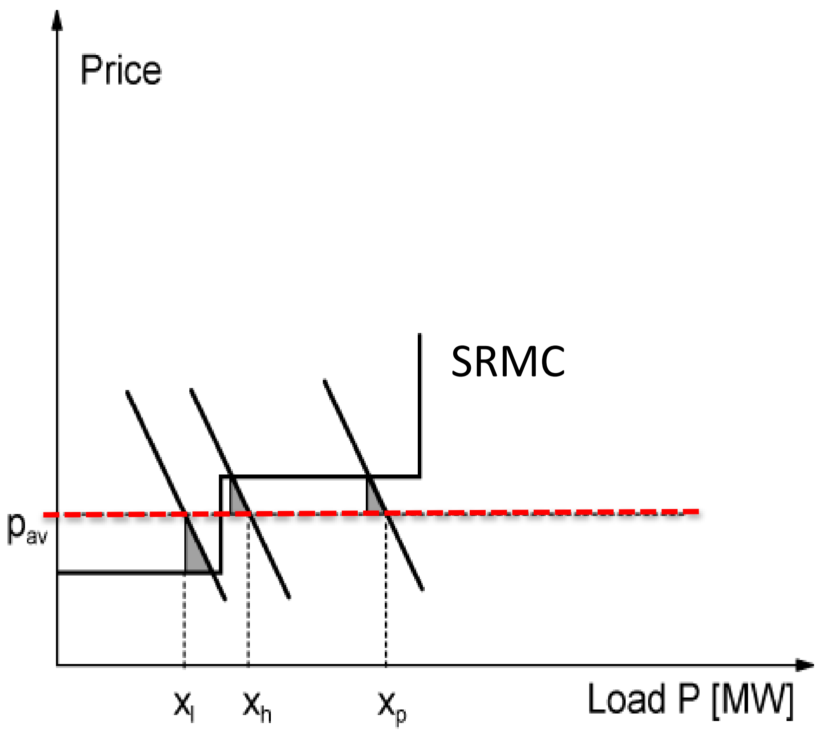
- Time-Of-Use (TOU):

$$C = c_{fixed} + \sum_{p=1}^{n} c_{en,p} \cdot Q \text{ [NOK]}$$

  - p is period of day, n is number of periods (typical 2-4), $c_{en,p}$ is energy rate in each period
  - very common in many countries (Europe, US); "day/night tariff"



Impact TOU rates



- **Three periods, l, h, p**
- **Average price results in loss of social welfare**: pricing does not reflect the actual cost of electricity generation during different demand periods
- **Pricing according to SRMC (short-run marginal cost) increases SW with the sum of shaded triangles**: ensures that prices reflect the actual cost of producing electricity at a given time
- **Assuming TOU rates exactly reflect SRMC…**

Spot pricing

Fred Schweppe, "Saint Fred", 1980's

$\rho_i(t) =$	$\lambda(t)$ $+ \ \gamma_Q(t)$ $+ \ \eta_{L,i}(t)$ $+ \ \eta_{Q,i}(t)$ $+ \ \eta_{R,i}(t)$	**[System Lambda]** **[Generation quality of supply]** **[Marginal network losses]** **[Network quality of supply]** **[Revenue Reconciliation]**

Schweppe, F.C., Caramanis, M.C., Tabors, R.D., Bohn, R.E., “Spot Pricing of Electricity”, (Kluwer Academic Publishers, Boston, MA, 1988)

Norway: typical household and small business contracts

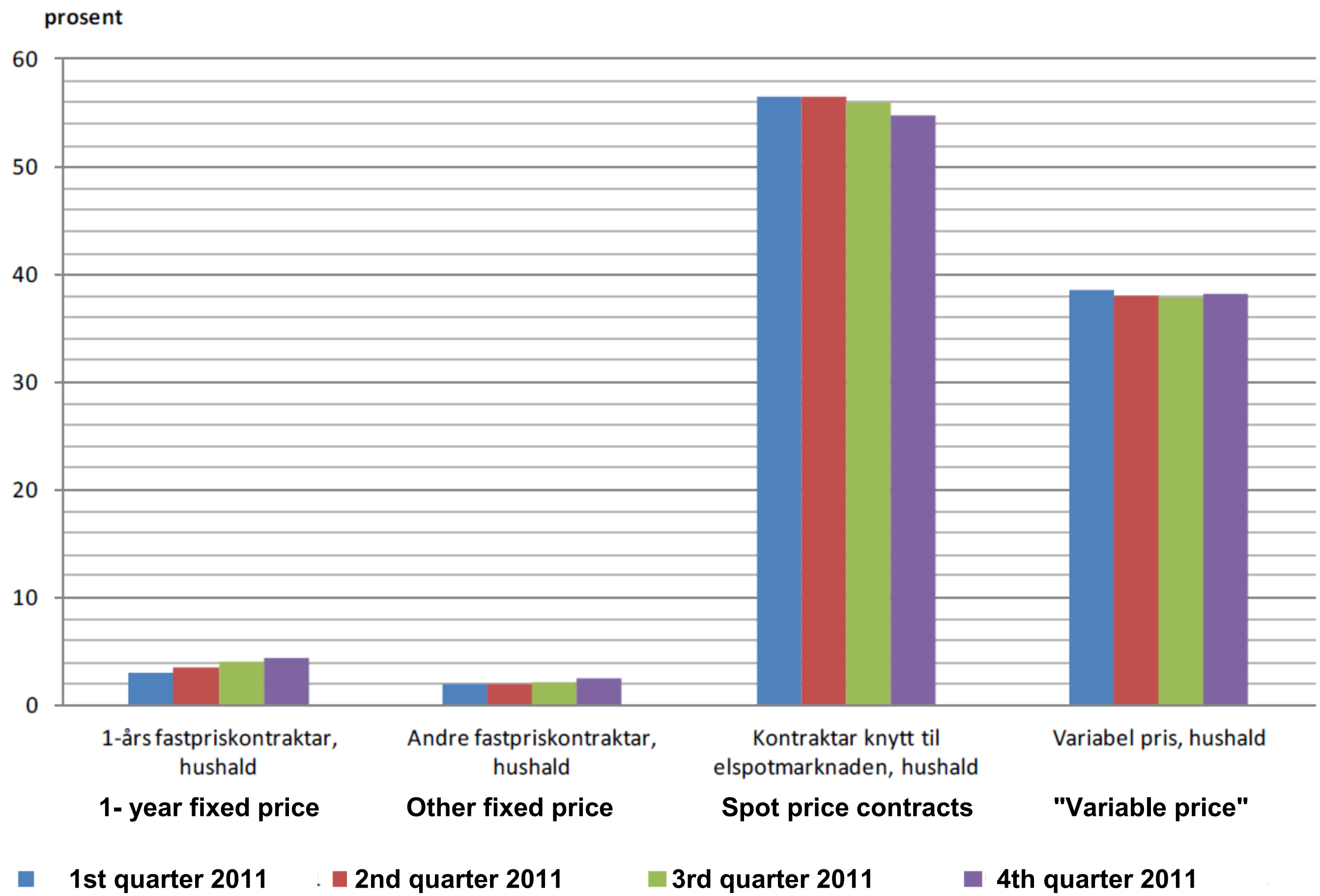
- Fixed Price Contracts (Fastpris):
  - Fixed energy rate, 6 months,12 months and more
- Spot Price Contracts (Spotpris):
  - price follows spot price + add on (= profit for retailer)
  - hourly demand based on the recorded profile (AMS)
- Plus Agreements(Plussavtale),
  - is typically an agreement for customers who both consume and produce electricity, for example, from solar panels and can sell excess power back to the grid or power supplier.

- Government Support Schemes (Norgespris & Strømstøtte):
  - Strømstøtte: automatic electricity price compensation for households when spot prices exceed a defined threshold, covering a share of consumption
  - Norgespris: optional state-backed fixed-price scheme offering households a predictable electricity price over time, reducing exposure to high and volatile spot prices

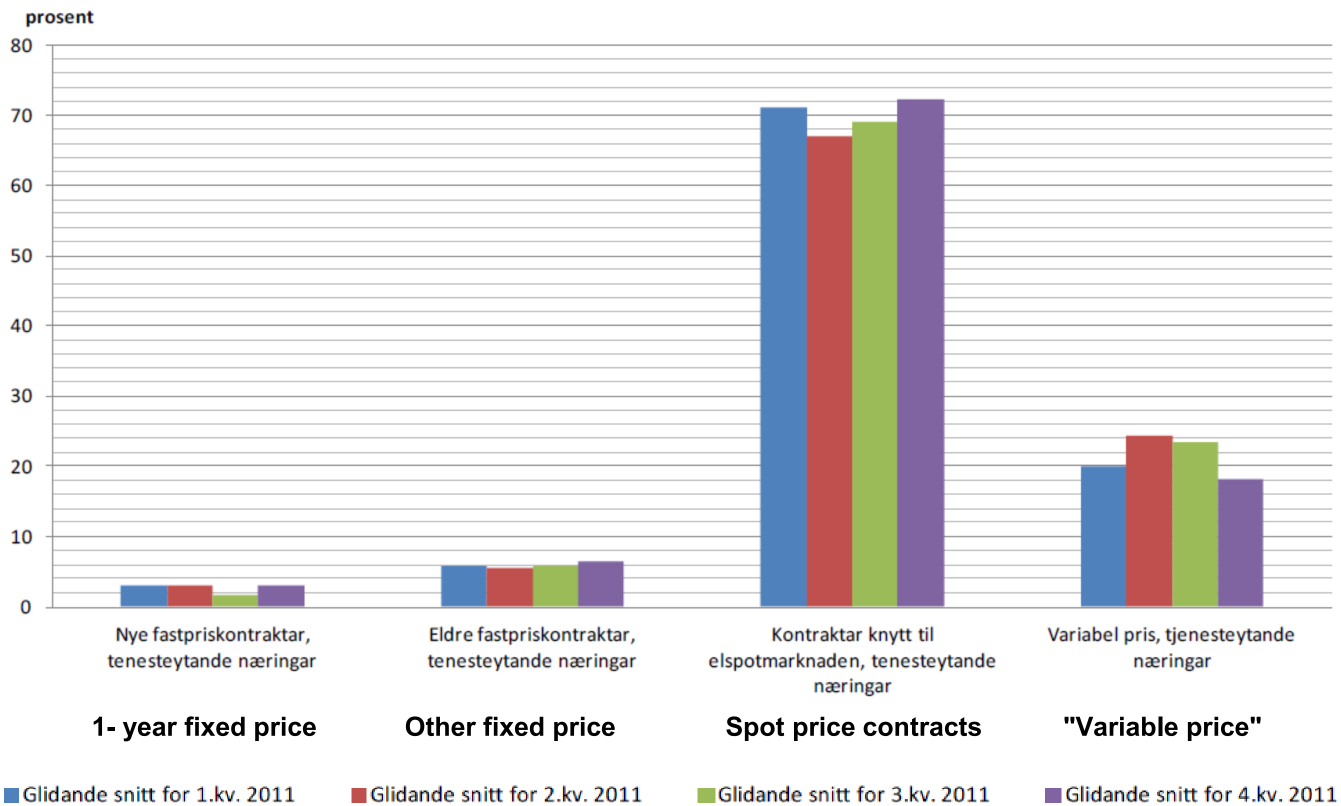
https://www.strompris.no/#/?gridId=249&gridName=TRØNDERENERGI+NETT+AS&municipalityId=363

The Norwegian Energy Regulatory Authority, RME Rapport, Nr 8/2022: https://publikasjoner.nve.no/rme_rapport/2022/rme_rapport2022_08.pdf

Market share for different contract types, <u>households</u>

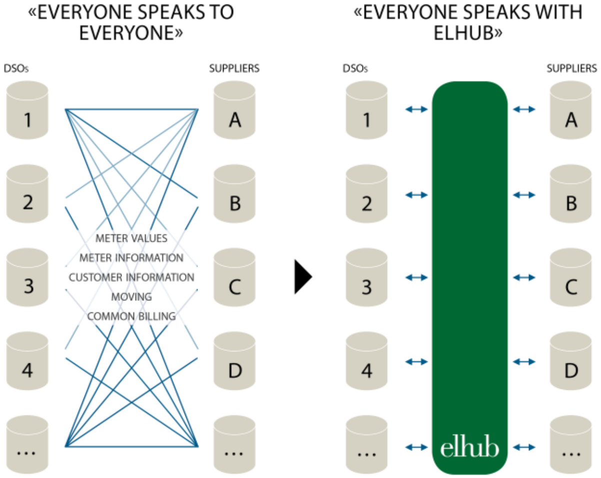


Market share for different contract types, service sector



Advanced Metering Infrastructure

- In Norway, an advanced metering infrastructure (AMI) for hourly meter readings has been established for all households.
- The AMI rollout was completed by the end of 2019, as required by the Norwegian Water Resources and Energy Directorate (NVE)
- Elhub was put into operation by Statnett in February 2019
  - facilitates the automation and digitalization of value chains related to the exchange of meter data
  - changes in suppliers
  - sharing of customer information
  - financial settlements



https://elhub.no/om-elhub/bli-kjent-med-elhub/
https://youtu.be/uYVo1MHBFvc?feature=shared

Marginal cost pricing

- Spot pricing in Norway is "almost" marginal cost pricing
- Theoretically attractive, but
  - More volatile, more complicated
  - Politically hard sell
- Average cost pricing is easier to understand. But inefficient!

- MC pricing is hardly understood and even less appreciated outside the economics profession
  - cf. media focus each time Norwegian power prices are "high"
- Is seen as "unfair"
  - cost is low, price is high

Strømprisene

Regjeringen: Folk kan velge mellom strømstøtte og fastpris på 40 øre

Regjeringen legger frem flere nye strømgrep, inkludert en ordning med fastpris på 40 øre kilowattimen for hytteeiere og husholdninger.

Analytiker advarer mot fastpris: – Ekstreme priser

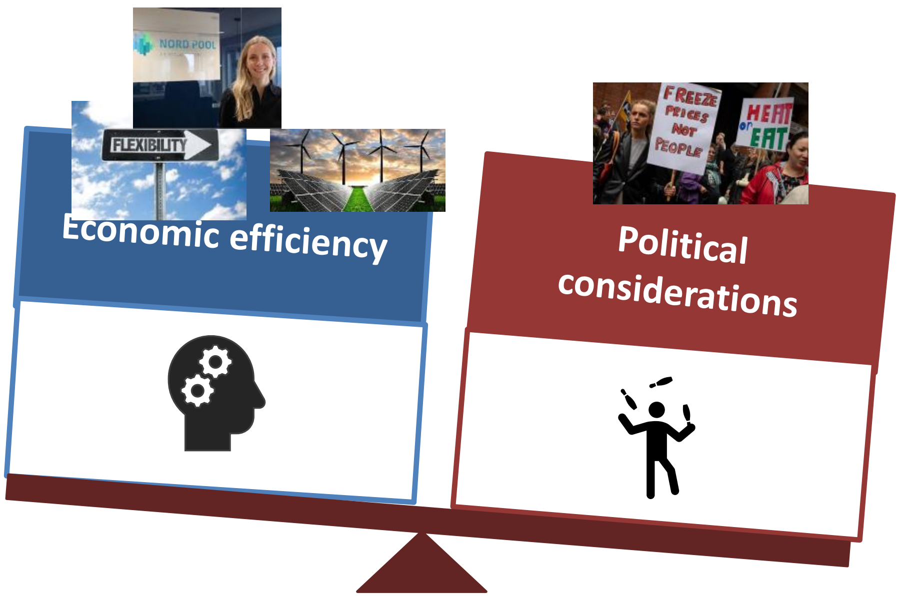
Avtalen er gunstig, men neppe bærekraftig, ifølge analytiker.

Gjestad trafostasjon og høyspentledninger i Jessheim. Foto: Marius Helge Larsen / NTB

Norgespris

- Government proposal: Norgespris (fixed 50 øre/kWh including VAT)
- Consumption caps:
  - 5,000 kWh/month for households
  - 1,000 kWh/month for cabins
- Duration: Oct 2025 – Dec 2026
  - Valid for consumption within the cap
    - Usage above cap billed at market price
    - Standard grid tariffs & surcharges still apply
- Aim: predictability & security for households.

Political will is not always aligned with economic efficiency



Disconnects the price from the value of electricity

Core Weaknesses of Norgespris

- Distributional injustice: the largest users get the most support.
- Efficiency loss: removes incentives for:
  - Energy-saving investments
  - Demand flexibility (load shifting)
  - Solar/self-production
  - Grid Expansion
- Risks higher market prices & harms industry competitiveness.

“Emergency intervention" of the electricity market.

- Energy crisis: Ursula von der Leyen (president of the European Commission) calls for 'emergency intervention' in electricity market

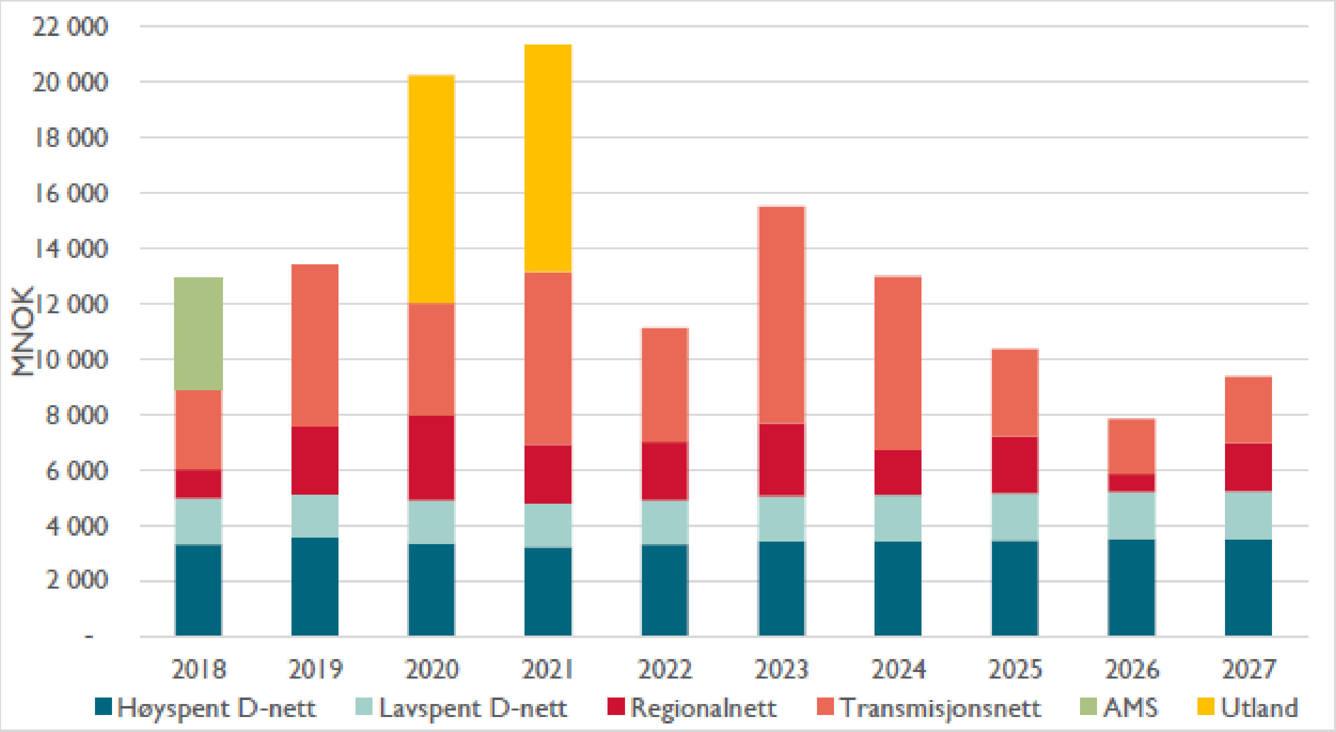
- Long-term contracts as the backbone
    - Expansion of Contracts for Difference (CfDs) for low-carbon generation (RES & nuclear)
    - Stronger role for Power Purchase Agreements (PPAs)
- Improved consumer protection
    - Fixed-price and long-term retail contracts encouraged
    - Emergency retail price interventions allowed under strict conditions
- Stronger incentives for:
    - Demand response
    - Storage
    - Capacity mechanisms (with tighter EU coordination”)

https://energy.ec.europa.eu/topics/markets-and-consumers/electricity-market-design_en?utm_source#reform-of-the-electricity-market-design

GRID TARIFFS

Total expected investments in the grid (2018-2027)

- 135 billion NOK is distributed at the various grid levels



Figur 26: Totale investeringskostnader i Norges strømnett per år, sortert etter idriftsettelsesår. For regionalnettet er kun de scenariouavhengige investeringene inkludert.

Retail distribution is almost the most costly part of transmission/distribution

Source: NVE (https://publikasjoner.nve.no/rapport/2018/rapport2018_103.pdf)

Why are grid tariffs important?

DSO

Varsel om inntektsrammer for 2022 eksl Statnett

19,3 mrd.

Investeringer og høyere rente gir større kapitalkostnader

Økte kraftpriser gir større nettapskostnader

+

TSO

Varsel om inntektsrammer for 2022 for Statnett

9,9 mrd.

Investeringer og høyere rente gir større kapitalkostnader

Økning i nettaps- og systemansvarskostnader

Flaskehalsinntekter vil dempe tarifføkning

=

Varsel om inntektsrammer for 2022 - samlet

29,2 mrd.

Anmodet nettselskaper med merinntekt til å utbetale denne i 2022

Selskaper med mindreinntekt kan søke om dispensasjon fra å styre saldo mot null

Ingen sammenheng mellom økning i inntektsrammer og ny nettleiemodell

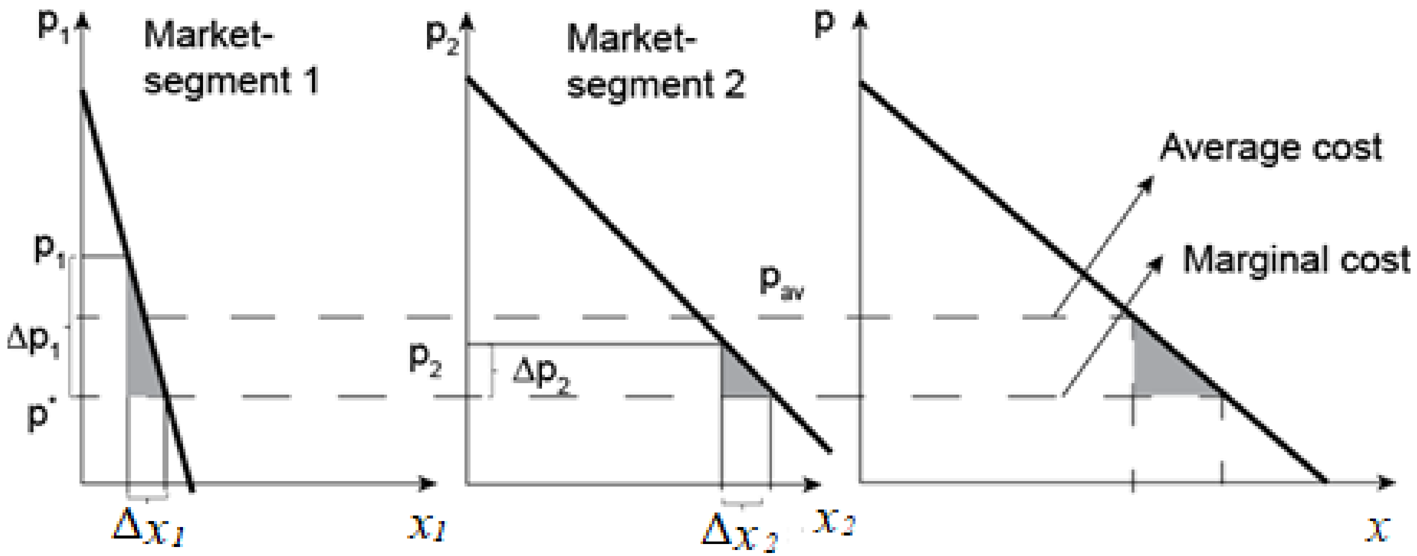
Grid Tariff Overview

- **What is a Grid Tariff?** A charge to consumers for the use of the electricity grid, covering costs for transmission, distribution, and maintenance.

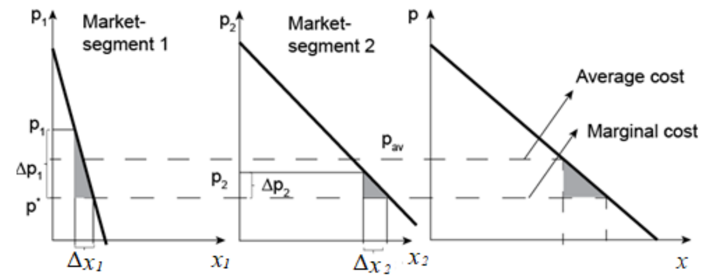
- **Components of a Grid Tariff:**
  - **Fixed Charges (NOK):** Costs independent of usage, ensuring grid availability.
  - **Variable Charges(NOK/kWh):** Based on energy consumption or peak demand.
  - **Capacity Charges (NOK/kW):** Reflect maximum load requirements.

Ramsey Pricing

The grid is not part of the competition; we need a grid to transfer power. We need to cover the cost of the grid and there is a huge fixed cost. How do we share this cost among various groups of consumers?→ Ramsey pricing (A theoretical foundation)



Ramsey Pricing



Market segment 1: $\Delta x_1 = -\Delta p_1 \frac{\partial x_1}{\partial p_1}$

Market segment 2: $\Delta x_2 = -\Delta p_2 \frac{\partial x_2}{\partial p_2}$

Welfare loss → The area of two triangles $Loss = \Delta p_1^2 \left(\left(-\frac{1}{2}\right) \frac{\partial x_1}{\partial p_1} \right) + \Delta p_2^2 \left(\left(-\frac{1}{2}\right) \frac{\partial x_2}{\partial p_2} \right)$

The objective is then to minimize the welfare loss subject to the condition (K is a constant revenue):

$$x_1 \Delta p_1 + x_2 \Delta p_2 = K$$

$$\text{Min} \;\; Loss = \Delta p_1^2 \left(\left(-\frac{1}{2}\right) \frac{\partial x_1}{\partial p_1} \right) + \Delta p_2^2 \left(\left(-\frac{1}{2}\right) \frac{\partial x_2}{\partial p_2} \right)$$

$$st.$$

$$x_1 \Delta p_1 + x_2 \Delta p_2 = K$$

We can find Δp_2 from the constraint and set in the objective function

Ramsey Pricing

We can find Δp_2 from the constraint

$$\Delta p_2 = \frac{K - x_1 \Delta p_1}{x_2} \qquad (1)$$

Set Δp_2 in the objective function

$$Loss = \Delta p_1^2 \left(-\frac{\partial x_1}{2\partial p_1} \right) + \left(\frac{K - x_1 \Delta p_1}{x_2} \right)^2 \left(-\frac{\partial x_2}{2\partial p_2} \right)$$

Minimum losses are found by setting the derivative with respect to Δp_1 equal to zero

$$\Delta p_1 \left(\frac{\partial x_1}{\partial p_1} \right) = \left(\frac{K - x_1 \Delta p_1}{x_2} \right) \left(\frac{x_1}{x_2} \right) \left(\frac{\partial x_2}{\partial p_2} \right) \qquad (2)$$

Combining Equations (1) and (2) and leads to:

$$\Delta p_1 \left(\frac{\partial x_1}{\partial p_1} \right) \left(\frac{1}{x_1} \right) = \Delta p_2 \left(\frac{\partial x_2}{\partial p_2} \right) \left(\frac{1}{x_2} \right) \qquad (3)$$

Ramsey Pricing

$$\left.\begin{aligned}
&\Delta p_1\left(\frac{\partial x_1}{\partial p_1}\right)\left(\frac{1}{x_1}\right)=\Delta p_2\left(\frac{\partial x_2}{\partial p_2}\right)\left(\frac{1}{x_2}\right)\\
&\Delta x_1=-\Delta p_1\frac{\partial x_1}{\partial p_1}\\
&\Delta x_2=-\Delta p_2\frac{\partial x_2}{\partial p_2}
\end{aligned}\right\}\Rightarrow\frac{\Delta x_1}{x_1}=\frac{\Delta x_2}{x_2}$$

The demand price elasticity ε defined as: p_2 $\quad \varepsilon(x)=-\left(\frac{\partial x}{\partial p}\right)\left(\frac{p}{x}\right)$

By rewriting Equation (3) in the following way $\left(\frac{\Delta p_1}{p_1}\right)\left(\frac{\partial x_1}{\partial p_1}\right)\left(\frac{p_1}{x_1}\right)=\left(\frac{\Delta p_2}{p_2}\right)\left(\frac{\partial x_2}{\partial p_2}\right)\left(\frac{p_2}{x_2}\right)$

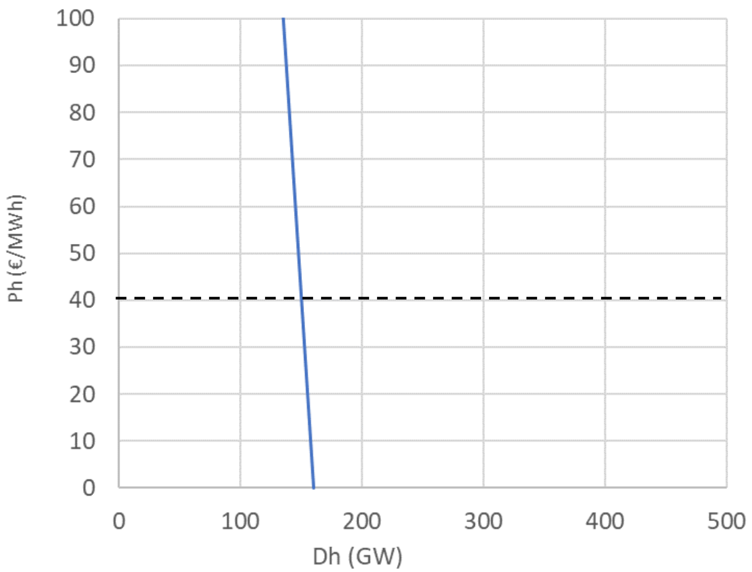
We get **Inverse Elasticity Rule**

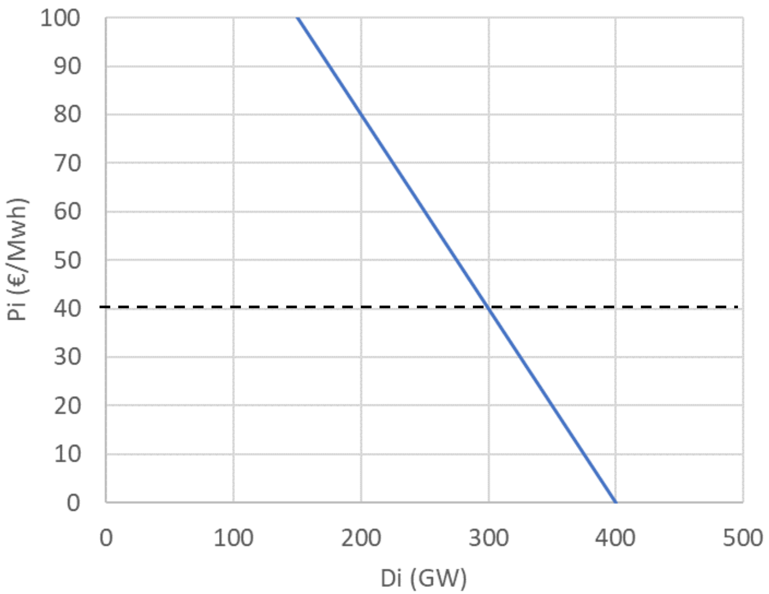
$$\boxed{\frac{\frac{\Delta p_1}{p_1}}{\frac{\Delta p_2}{p_2}}=\frac{\varepsilon_2}{\varepsilon_1}}$$

Inverse elasticity rule, says that the deviation in price relative to the optimum (which is usually marginal cost) shall be proportional to the inverse of the elasticity.

Example Ramsey pricing

- Household demand (in GWh): $p_h = 640 - 10^{-3} \times (4\ D_h)$
- Industry demand (in GWh): $p_i = 160 - 10^{-3} \times (0.4\ D_i)$





Ramsey pricing, exercise

- **The (average) power price is 40 €/MWh**
- **The grid company needs annual revenues of 3 million Euro**
- **The revenues are obtained by adding a grid tariff to the power price**

➢ Calculate demand in both sectors if there was no grid tariff

➢ What are prices and demand if **Ramsey pricing** is used to determine the grid tariff?

➢ What are prices and demand if both sectors pay the **same tariff**?

- *Tips:*
  - *Set up the equations that determine the solution, check that #eq = #var*
  - *Demand is given in GWh and prices in €/MWh, so take into account the factor 1000 in your equations!*

- Demand in each sector
  - set in p=40 in demand functions, D_h=150 GWh, D_i=300 GWh

- **Ramsey pricing**: set up the equations below
  - 4 unknowns: Δp_h, Δp_i, D_h, D_i
  - 4 equations

Ramsey pricing (eq. 4.10 in Book): $$\Delta p_h \cdot \frac{\partial D_h}{\partial p_h} \cdot \frac{1}{D_h} = \Delta p_i \cdot \frac{\partial D_i}{\partial p_i} \cdot \frac{1}{D_i}$$

Revenue requirement: $$\Delta p_h \cdot D_h + \Delta p_i \cdot D_i = R \cdot 10^{-3}$$

Demand equations: $$p_h = p + \Delta p_h = a_h - b_h D_h$$
$$p_i = p + \Delta p_i = a_i - b_i D_i$$

The demand equations can of course be rewritten to give D as a function of p + Δp

Setting in the expressions for D in the equation for Ramsey pricing, you can express Δp_i as a function of Δp_h:

$$\Delta p_i = \Delta p_h \cdot \frac{a_i - p}{a_h - p} = \Delta p_h \cdot d$$

Setting this in the equation for the revenue requirement gives after some reordering:

$$\Delta p_h \cdot \frac{a_h - (p + \Delta p_h)}{b_h} + \Delta p_h \cdot d \cdot \frac{a_i - (p + d \cdot \Delta p_h)}{b_i} = R \cdot 10^{-3}$$

or with some reordering:

$$-(\frac{1}{b_h} + \frac{d^2}{b_i})\Delta p_h^2 + \left(\frac{a_h - p}{b_h} + d \cdot \frac{a_i - p}{b_i} \right)\Delta p_h - R \cdot 10^{-3} = 0$$

before solving, set in values:

$$-0.35\Delta p_h^2 + 210\Delta p_h - 3000 = 0$$

Solving gives the following result:

$$\Delta p_h = 14.64 \quad \vee \quad 300.00$$
$$\Delta p_i = 2.93 \quad \vee \quad 60.00$$

Both solutions are mathematically valid, but obviously the lower solution gives higher social welfare and satisfies the revenue requirement.

solution values are:

$$D_h = 146.3 \text{ GWh, reduction } 2.44\ \% \leftarrow (D_h = 150 \text{ GWh})$$
$$D_i = 292.7 \text{ GWh, reduction } 2.44\ \% \leftarrow (D_i = 300 \text{ GWh})$$
$$R = (146.3 \cdot 14.64 + 292.7 \cdot 2.93) \cdot 10^3 = 3 \text{ million €}$$

With the same tariff Δp for both sectors, the equation for Ramsey pricing is not used, and the following equations emerge:

- 3 unknowns: Δp, D_h, D_i
- 3 equations

Revenue requirement:	$\Delta p \cdot D_h + \Delta p \cdot D_i = R \cdot 10^{-3}$
Demand equations:	$p + \Delta p = a_h - b_h D_h$
	$p + \Delta p = a_i - b_i D_i$

Expressing D as a function of Δp and setting in in the revenue equation gives after similar reordering as before:

$$-(\frac{1}{b_h} + \frac{1}{b_i})\Delta p^2 + \left(\frac{a_h - p}{b_h} + \frac{a_i - p}{b_i}\right)\Delta p - R \cdot 10^{-3} = 0$$

before solving, set in values:

$$-2.75\Delta p_h^2 + 450\Delta p_h - 3000 = 0$$

Solving gives the following result:

$$\Delta p = 6.96 \;\vee\; 81.82$$

Again both solutions are mathematically valid, but like before the lower solution gives higher social welfare and satisfies the revenue requirement .

Solution values are:

$$D_h = 148.3 \text{ GWh, reduction } 1.16\ \% \leftarrow (D_h = 150 \text{ GWh})$$
$$D_i = 282.6 \text{ GWh, reduction } 5.8\ \% \leftarrow (D_i = 300 \text{ GWh})$$
$$R = (148.3 + 282.6) \cdot 6.96 \cdot 10^3 = 3 \text{ million €}$$

If you want, you can calculate the loss in social welfare for both solutions and compare

Benefits of Ramsey Pricing in Grid Tariffs

- Encourages efficient grid utilization.
- Supports cost recovery for grid operators.
- Reduces economic distortions in electricity use.
- Provides a framework for integrating demand-side flexibility.

Challenges in Implementation

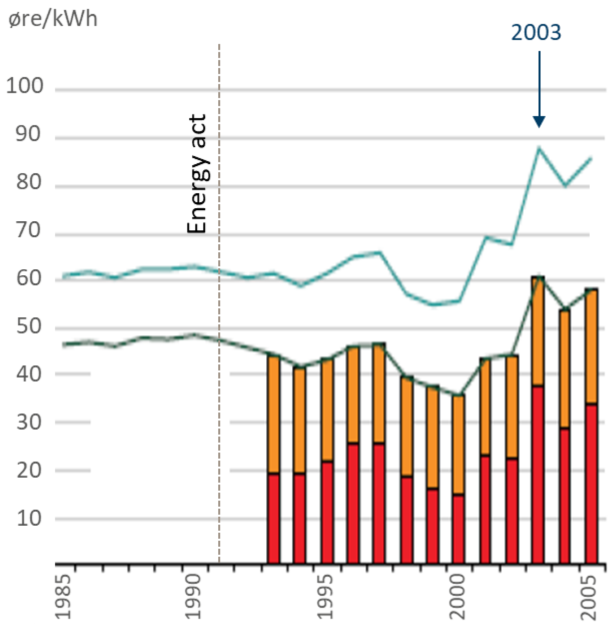
- **Estimating Elasticities**:
  - Accurate data on consumer behavior is critical.
- **Equity Concerns:**
  - Higher tariffs for inelastic consumers (e.g., households) may be seen as unfair.
- **Dynamic Grid Conditions:**
  - Integration of renewables and EVs complicates cost and demand predictions.

Norwegian Grid Tariff

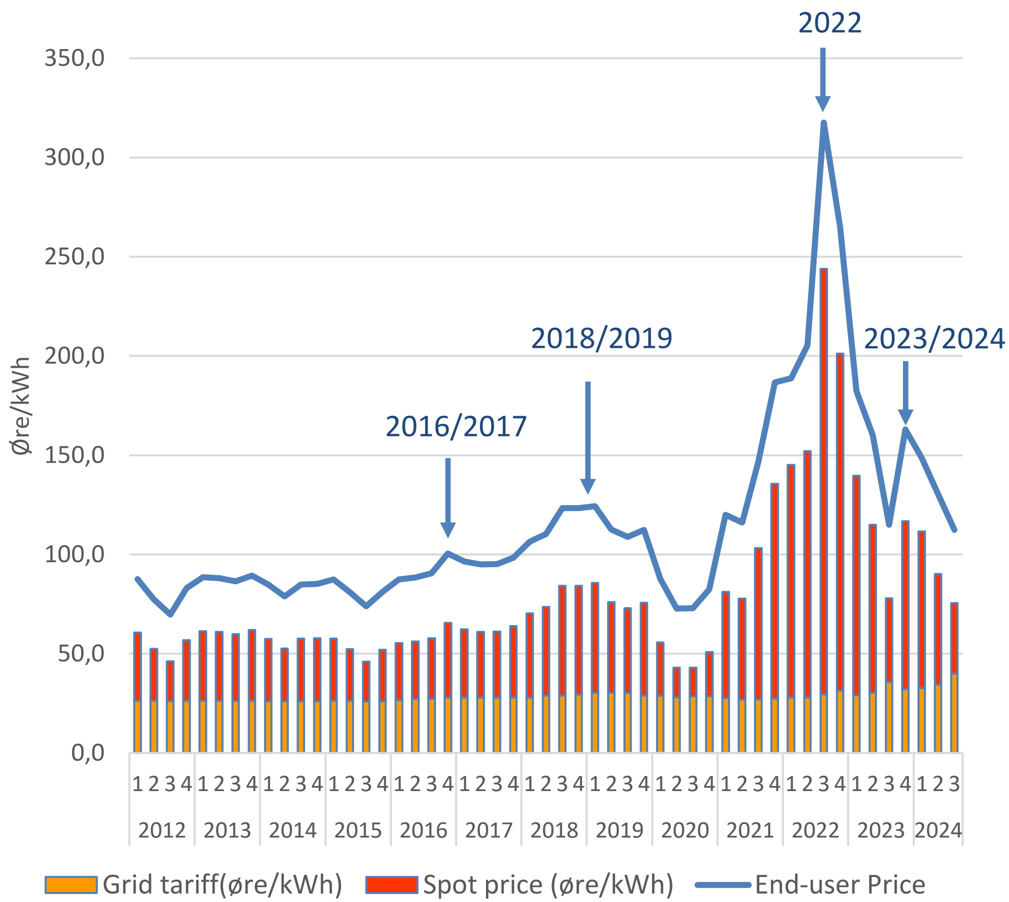
- **Current Structure:**
  - Fixed Component: A monthly charge covering grid connection and basic infrastructure costs.
  - Energy-Based Component: Based on the total kWh consumed, reflecting usage levels.
  - Peak Demand Component kW (average consumption during three monthly peaks): Introduced to encourage demand-side management and reduce grid stress during peak hours.
- **Key Features:**
  - Incentivizes energy efficiency and demand response.
  - Supports integration of renewable energy by reducing peak demand volatility.
- **Recent Reforms (July 2022):**
  - Transition towards more dynamic and capacity-based pricing.
  - Focus on aligning tariffs with actual grid costs and usage patterns.
- **Challenges:**
  - Public acceptance of peak demand charges.
  - Ensuring fairness across urban and rural consumers.
  - Managing the impact on low-income households and small businesses.

Norwegian household's electricity price (quarterly)

- Households
  - Spot price
  - Grid tariff
  - VAT (25%)



Source: SSB/NVE, 2020
* Average prices for the whole country. Local differences exist.



Source: Norwegian Water Resources and Energy Directorate.
https://www.ssb.no/en/statbank/table/09387/tableViewLayout1/

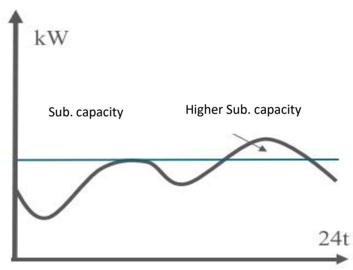
The suggestion from the Norwegian Regulator?

- The general idea was to move from **energy-based** grid tariffs to **capacity-based** grid tariffs→ Encouraging demand-side management

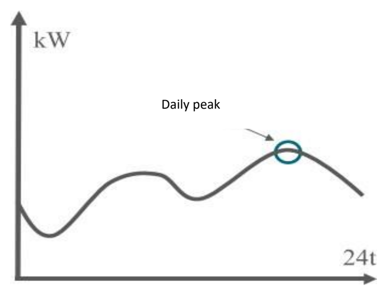
- Energy based:
  - Flat rate
  - Time-of-use
  - Peak/off-peak

- Capacity based:
  - Subscribed capacity
  - Demand charges (measure peak)

Subscription tariff



Daily peak load tariff



Sigurd Bjarghov, Hossein Farahmand, Gerard Doorman, "Capacity subscription grid tariff efficiency and the impact of uncertainty on the subscribed level", Energy Policy, Volume 165, 2022

Grid tariffs cause drama?

Effektbasert nettleie: Arroganse og skrivebords-bestemmelser

Capacity-based grid tariffs: Arrogance and useless regulations

the proposal for new grid tariff is a big scam and should be sent back on the desk

Forslaget om ny nettleie er en stor bløff og bør sendes tilbake til skrivebordet

Linda Ørstavik Öberg

29. september 2020

the new grid tariff solves nothing

Innlegg: Ny nettleie løser ingenting

The new grid tariff is too complicated and rewards those who waste electricity

If NVE gets what they want, there will be eight years of chaos in electricity bills

Asks the government to drop NVE's proposal for grid tariff

Can the politicians stop NVE's grid tariff madness now

Kun politikerne kan stoppe NVEs nettleiegalskap nå!

The capacity-based tariff makes solar energy, heat pumps, and additional insulation become even more expensive for Norwegian households and will put an end to those who want to live more climate-friendly.

New tariff design in Norway

consumption below 100 000 kWh in one year
(from July 1, 2022)

Level (1-10)	Capacity (kW)	Capacity share (NOK/month)	Energy share Day (06:00-22:00) (øre/kWh)	Energy share Night/Weekend (øre/kWh)
Level 1	0-2 kW	125	43.10	36.85
Level 2	2-5 kW	200	43.10	36.85
Level 3	5-10 kW	325	43.10	36.85
Level 4	10-15 kW	450	43.10	36.85
Level 5	15-20 kW	575	43.10	36.85
Level 6	20-25 kW	700	43.10	36.85
Level 7	25-50 kW	1 325	43.10	36.85
Level 8	50 – 75 kW	1 950	43.10	36.85
Level 9	75 – 100 kW	2 575	43.10	36.85
Level 10	Over 100 kW	5 150	43.10	36.85

Roman J. Hennig, David Ribó-Pérez, Laurens J. de Vries, Simon H. Tindemans, «What is a good distribution network tariff?—Developing indicators for performance assessment», Applied Energy, Volume 318, 2022.

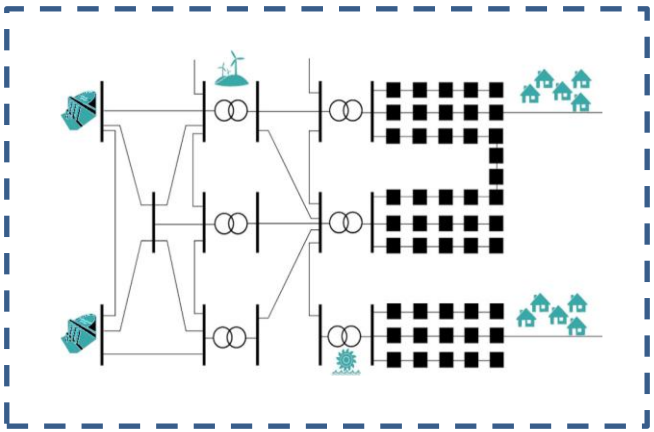
The future energy system

- Renewable power production
  - Instant consumption and generation
  - Peak load is dimentioning for the electricity grid
  - **→ consumption must adapt to the production and the capacity of the grid**

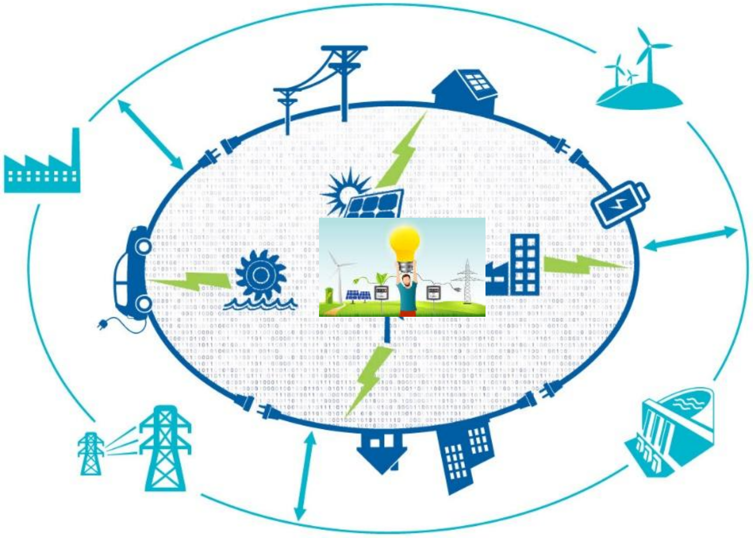
- Buildings become an integrated part of the future energy system
  - Produce and consume electricity → PROSUMERS
    - **FLEXIBLE DEMAND**
    - **DEMAND RESPONSE**
    - **DEMAND SIDE MANAGEMENT**

The transition strategy towards a more decentralized and consumer-centric model

Power system of today



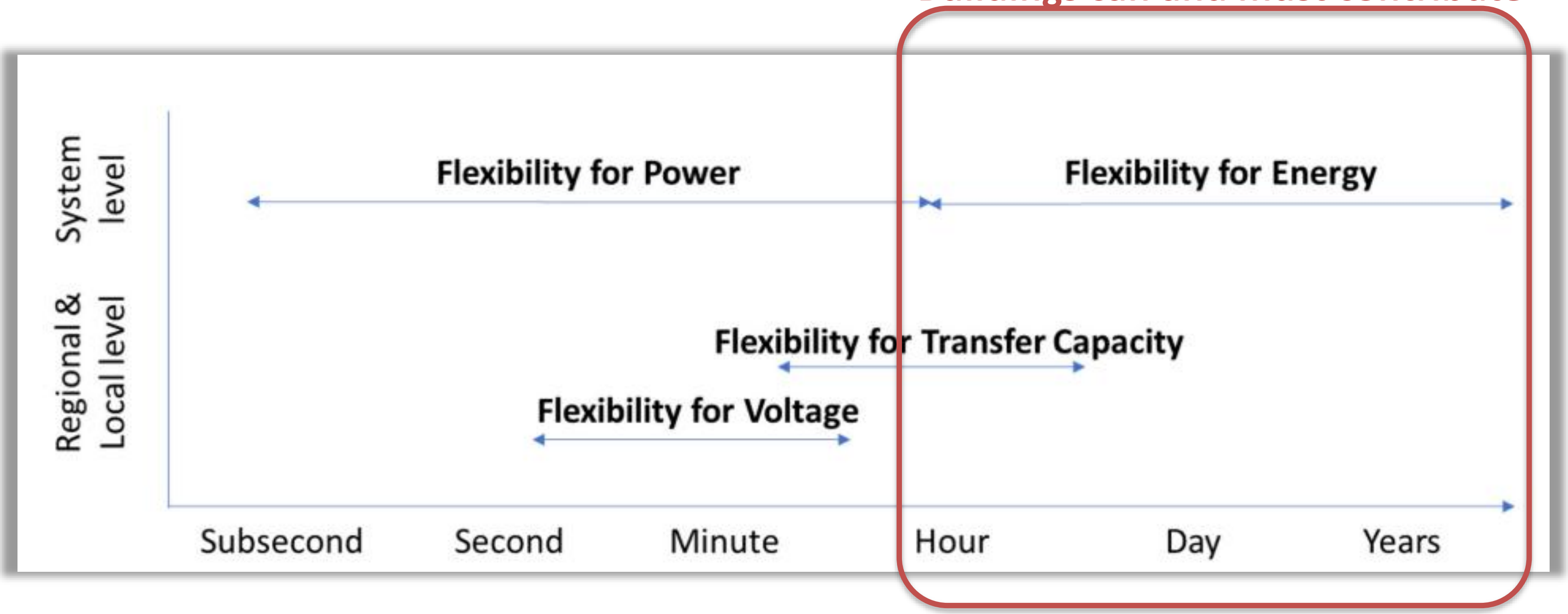
The future power system



Distribution grids are substantially evolving to adapt to this new reality

Flexibility needs in the power system

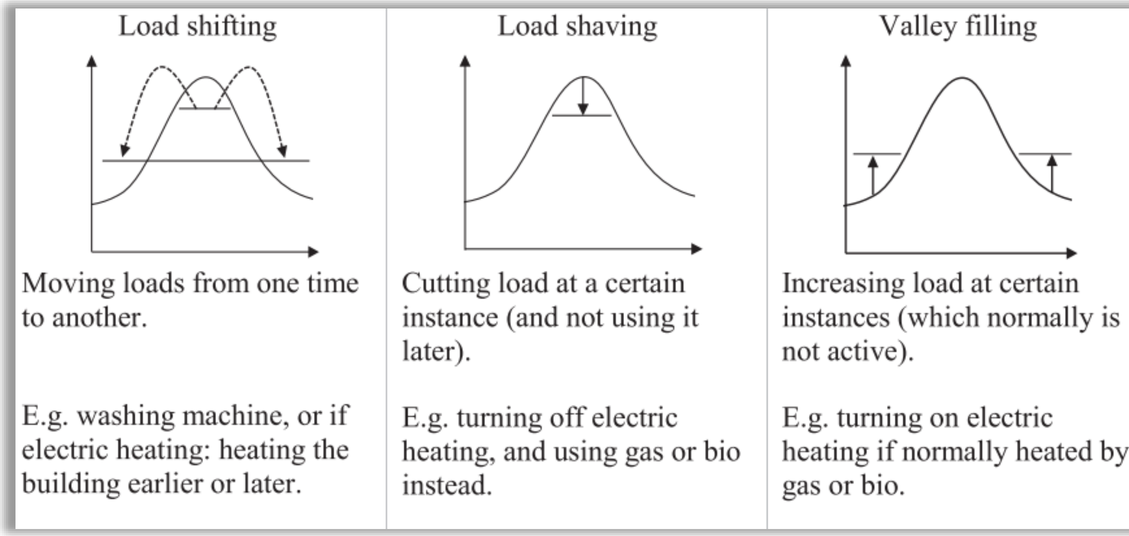
Buildings can and must contribute



(IEA ISGAN, 2019) Flexibility needs in the future power system, http://www.iea-isgan.org/flexibility-in-future-power-systems/

Energy Flexible Buildings

- Changing consumption from what it *would have been*
  - Flexible demand
  - Demand side management
  - Demand response (explicit and implicit)
    - Price-based
    - Incentive-based[1]
- How?
  - Moving loads in time
  - Shaving loads
  - Increasing loads



Source: Lindberg (2017), PhD thesis NTNU, http://hdl.handle.net/11250/2450566

NTNU

[1] Kasper Emil Thorvaldsen, PhD thesis, A Long-term Strategy Framework for Flexible Energy Operation of Residential Buildings: https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/3030494

Flexibility in an apartment block

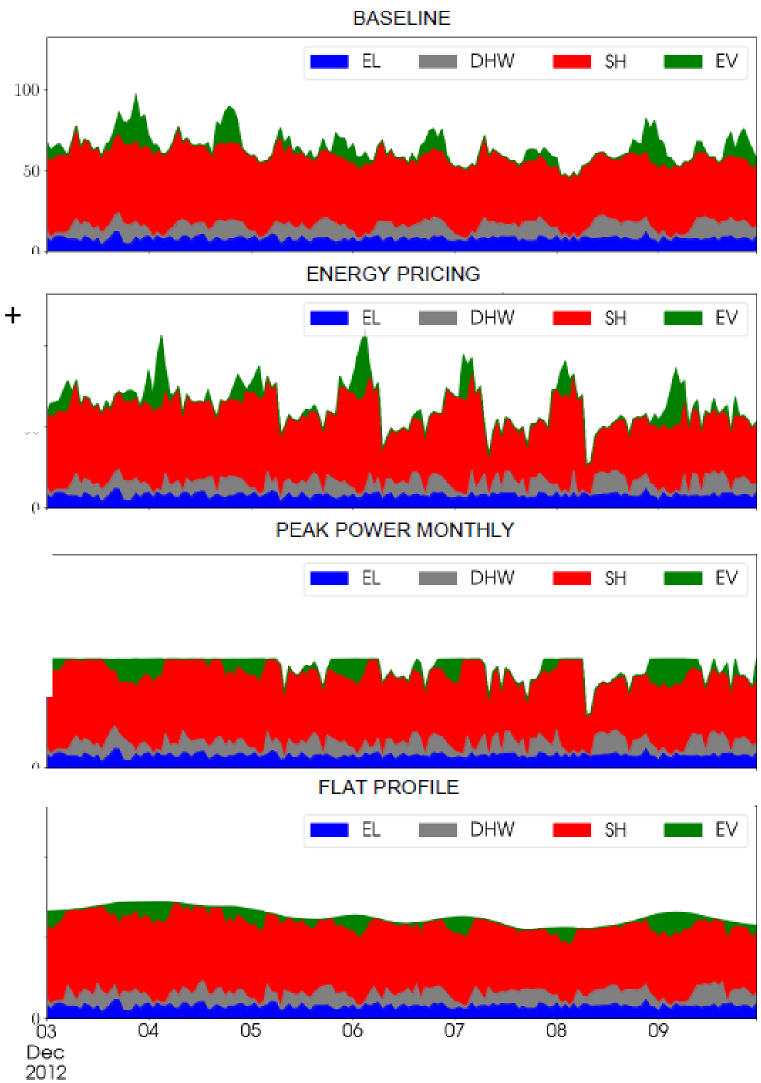
- Price-incentive decides the potential for
  - Load shifting
  - Peak load reduction
- Price-incentive consists of
  - Combination of spot price and grid tariff
  - New: flexibility markets?

Baseline (without flexibility)

Energy grid tariff + spot price

Monthly peak tariff + spot price

Flat load profile (regardless of price)



FLEXBUILD

SINTEF

Kilde: SINTEFs beregninger i FlexBuild-p

https://www.sintef.no/projectweb/flexbuild/

Demand response

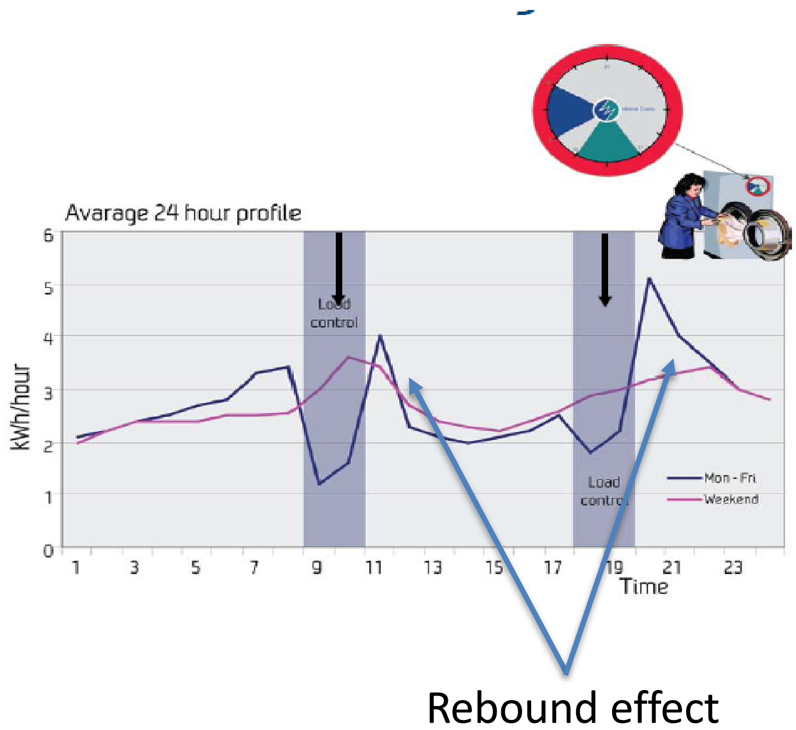
direct control of domestic hot water tanks (DHW)

Source: SINTEF Energi

Results:

Automatic Demand Response provides a stable accumulated demand response of 600 MW in peak hour

Load shifting of water heating the most efficient DR action.
Positive response from customers.
No cold water complaints.



https://smartgrids.no/app/uploads/2024/08/BattFLEX.pdf